

How to do the Skimming of the Data.

`Skimpy` is a Python library designed for data summarization and exploration in an efficient and visually appealing manner. It provides a concise yet informative overview of datasets, making it a useful tool for data scientists and analysts who need quick insights into their data.

Why is Skimpy Created?

Skimpy was developed to simplify exploratory data analysis (EDA) by summarizing datasets in a human-readable format. Traditional methods like Pandas' `.info()` and `.describe()` functions provide useful but sometimes hard-to-read outputs. Skimpy enhances readability by using color formatting and better structuring of data summaries.

The main goals of Skimpy are:

- Providing quick insights into datasets.
- Making summaries more readable through color-coded tables.
- Supporting efficient data exploration without excessive manual coding.

How Does Skimpy Work?

Skimpy works by analyzing Pandas DataFrames and summarizing key statistics for each column, including:

- Data type
- Number of missing values
- Number of unique values
- Most frequent values
- Mean, median, and standard deviation (for numerical columns)

It leverages `rich` (a Python library for colorful terminal output) to enhance readability.

Functions in Skimpy Library

Skimpy provides a few essential functions for summarizing and cleaning data.

A. `skim()` Function

- **Purpose:** Provides a detailed summary of a Pandas DataFrame.

B. `clean_columns()` Function

- **Purpose** : Cleans column names by standardizing them (e.g., removing spaces and special characters).

```
In [1]: # import the necessary libraries
import pandas as pd
import numpy as np
import skimpy as sp
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: # Load the titanic dataset that is a built in dataset in seaborn
titanic = sns.load_dataset('titanic')
titanic.head()
```

```
Out[2]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T

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```
In [3]: from skimpy import skim
# generate the summary statistics of the dataset
summary_stats = skim(titanic)
summary_stats
```

Data Summary

Data Types

Categories

Dataframe	Values	Column Type	Count	Categorical Variat
Number of rows	891	string	5	class
Number of columns	15	int64	4	deck
		float64	2	
		category	2	
		bool	2	

number

column	NA	NA %	mean	sd	p0	p25
survived	0	0	0.3838	0.4866	0	
pclass	0	0	2.309	0.8361	1	
age	177	19.865319865319865	29.7	14.53	0.42	20.
sibsp	0	0	0.523	1.103	0	
parch	0	0	0.3816	0.8061	0	
fare	0	0	32.2	49.69	0	7.

category

column	NA	NA %	c
class	0	0	F
deck	688	77.21661054994388	F

bool

column	true	true rate
adult_male	537	
alone	537	

string

column	NA	NA %	shortest	longest	min	max
sex	0	0	male	female	female	male
embarked	2	0.2244668911335578	S	S	C	S
who	0	0	man	woman	child	woman
embark_to	2	0.2244668911335578	Cherbourg	Southampton	Cherbourg	Southampton
wn			g	on	g	on
alive	0	0	no	yes	no	yes